

MATRIX × AWS

Engineering dossier

Multi-agent orchestration for a Montpellier game

ENGLISH

Evidence edition · 2 October 2026

AWS

MODEL PROVIDER

AZURE

MODEL PROVIDER

GITHUB

MODEL PROVIDER

Independent project · Proposed AWS support pilot

For Choupi, companion in the making. Keep the useful work, the failures and the evidence.

01 / EXECUTIVE VIEW

Reading the evidence

MATRIX is a local production orchestrator for the Montpellier Godot project *De l'écran à l'abandon*, led by Mehdi Souissi. AWS is its historical model-provider layer; Azure and GitHub are being integrated without replacing the local game workspace or the producer's authority.

Publication status: documented prototype and integration record. This dossier is not a production certification, an independent security audit or an AWS endorsement.

Evidence class	How to read this dossier
OBSERVED	A supplied log or screenshot shows the event. This does not certify the entire system.
DECIDED	The producer explicitly chose a requirement or boundary. Implementation may remain incomplete.
PROPOSED	An engineering design, test or funding plan developed for this dossier; not an installed feature.
UNVERIFIED	The full source, raw test artifact or current account state is unavailable.

The full executable MATRIX source and `MATRIX_PROD_FINAL_REPORT.md` were not supplied as inspectable artifacts for this edition. Statements inside screenshots of agent reports are treated as reports, not independently reproduced results. No account is accessed and no model is invoked to produce this dossier.

Document route

03 Product · 04 Architecture · 05 Security · 06 AWS · 07 Azure · 08 GitHub/Codex · 09 Protocol incidents · 10 Agent contract · 11 Routing and memory · 12 Godot · 13 Audio · 14 Vision and sources · 15 Costs · 16 Operations · 17 Evidence plates · 18 Release gates · 19 AWS request · 20 Occitan edition · 21–22 References.

02 / PRODUCTION PURPOSE

A game, not a model showroom

The deliverable is a playable 3D Montpellier experience. The producer's concise brief — “je veux que montpellier ressemble a montpellier et que le jeu soit plus beau” — directs the agents toward recognisable places and useful changes rather than provider demonstrations. The book is the narrative source; the graphic adaptation and the game must not silently overwrite its canon.

Workstream	Expected production result
City and movement	Readable streets, trams and stops, coherent collision, navigation and performance in the actual Godot scene.
Story and exploration	A story mode grounded in the author's material and a free-exploration mode with intelligible interactions.
Subjective-experience mode	A distinct creative treatment, not a clinical simulator or a claim to represent every person's experience.
Sound	Prepared NPC dialogue, transport announcements, ambience, subtitles and optional controlled live conversation.

Scope boundaries

MATRIX coordinates development. It is not automatically a game-server backend, a payment service or a cryptocurrency system. Montpellier ECU and the wider creative ecosystem provide context, but token operations, financial transactions and mining are outside the requested AWS pilot.

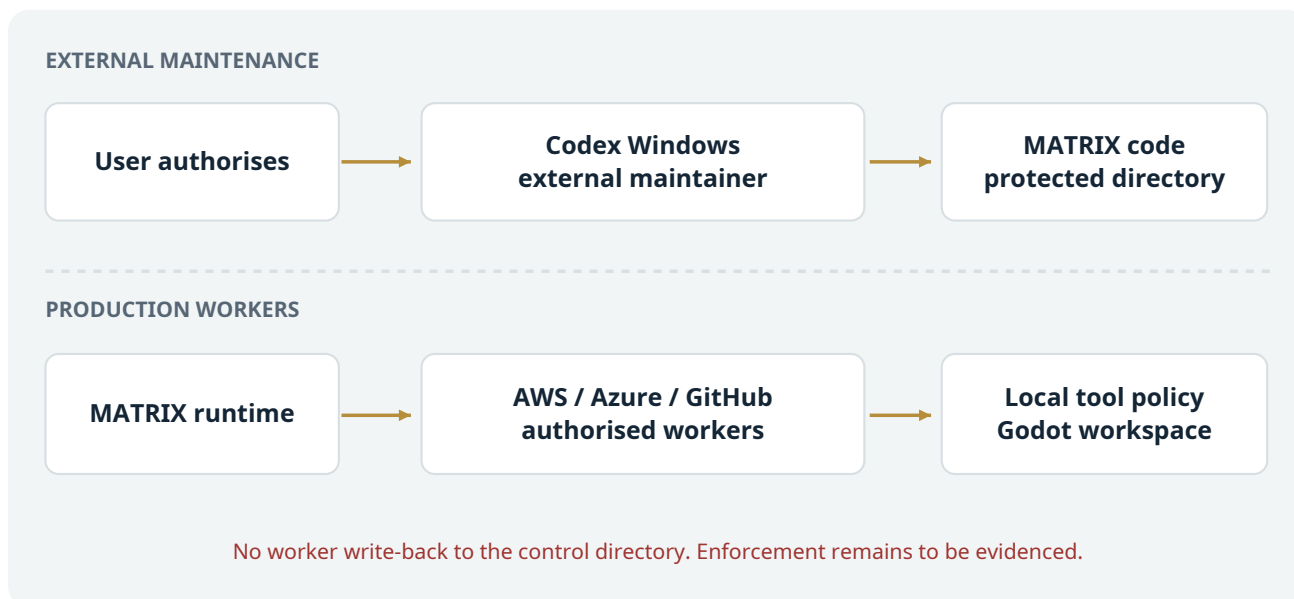
The protected control directory is D:\MTP-CENTRALE. The active game root must be resolved from the actual launcher and project configuration; older paths in conversations are not a safe deployment target. The external Codex maintenance environment and its proxy stay outside MATRIX.

Argent 0 as a production rule

The project philosophy is “L'argent n'est pas une fin. C'est un moyen.” For this dossier, its operational consequence is to reuse existing capacity, reduce redundant calls and buy evidence rather than model count. This is a producer decision, not a claim that cloud usage is free.

03 / ARCHITECTURE

Two paths, one human authority



The maintenance path and the production-worker path are different trust domains. The user starts Codex Windows as the external maintainer. A custom model provider changes how Codex obtains model responses; it does not move MATRIX's administration into a worker.

In the worker path, MATRIX selects an eligible agent, prepares authorised context, invokes a provider and receives a result. The local execution layer must enforce file and command boundaries before any proposed action reaches the game workspace.

What is actually observed

Screenshots show a local MATRIX console, a sequence of agent states and Codex Windows receiving text and requesting local tools through a custom provider. They do not expose every transport hop, operating-system identity or access-control rule. The diagram describes the intended separation, with the observed local paths identified.

Changing the Codex provider is not changing the maintainer. Launching a separate Copilot cloud agent against a remote repository is a different operation and did not recover the unpushed local MATRIX source.

Protecting the orchestrator

Seul le mainteneur externe explicitement lancé par l'utilisateur peut écrire dans D:\MTP-CENTRALE. MATRIX et tous les modèles qu'elle invoque n'ont jamais cette permission.

The quotation is the producer's authoritative security requirement. A denial string in a log, a prompt instruction and a UI permission label are not proof that Windows prevents the write. The evidence available here does not establish operating-system isolation.

Actor	Control directory	Authorised game workspace
External maintainer	Write only when explicitly launched by the user	Maintenance scope selected by the user
MATRIX runtime	No writes, including configuration and self-updates	Only the configured execution interface
Worker / child process	No writes, direct or indirect	Allowlisted scope; no self-escalation
Web or document content	Untrusted input, never an authority	Read as evidence, not as executable instructions

Proposed enforcement, not a deployment claim

Use distinct execution identities or a tested sandbox, restrictive ACLs, canonical-path checks and controlled child processes. The external maintainer must not hand its full token or arbitrary shell to workers. If all processes share the same unrestricted Windows identity, an application-only check is insufficient. [W11]

Place runtime logs, caches and mission state outside the protected tree. Otherwise MATRIX itself would violate the literal no-write rule. A proposed runtime-data directory is a design choice, not permission to migrate files automatically.

Acceptance must include direct write, rename, delete, symlink/junction traversal and a child-shell attempt against a harmless sentinel. All must fail under the worker identity, while an authorised game-workspace write succeeds. Record identity, resolved path, denial and before/after hash; never test by deleting real production files.



E09 · A maintainer approval prompt. Reading a denial rule is not proof that the worker identity is blocked by the operating system.

AWS as the specialist compute layer

The supplied MATRIX screens identify AWS-backed specialist profiles and show a production relay. Exact model IDs, regions, grants and live provider access are not verified for every profile. Display names such as Qwen, MiniMax, Kimi, Mistral and Claude must not be treated as an AWS availability catalogue. [E05–E06]

Proposed invocation path

mission → eligibility + policy → authorised context
→ AWS runtime adapter → model response
→ result normaliser → local tool policy → evidence

For supported models, Bedrock Converse/ConverseStream provides a common message interface and exposes stop reason, token usage and latency metadata. Model-specific capabilities must still be checked. These documented facilities support the proposed adapter; they do not prove that MATRIX currently calls Converse. [W01]

Concern	Implementation target
Identity	A least-privilege runtime role limited to approved inference resources. Keep deployment/IAM administration in a separate maintainer role.
Model mapping	Store provider, region, model ID or inference profile, supported input modes and contract version. Never route by display name alone.
Tools	Treat a model tool request as data. A local authorisation layer decides whether and where to execute it.
Reliability	Bound retries, honour throttling, preserve partial output, and open a circuit breaker after repeated provider failures.
Evidence	Record mission/attempt ID, resolved model, provider request ID, stop reason, usage and validation result without secrets.

Logging boundaries

CloudTrail and model-invocation logging answer different questions. Invocation logging can include full inputs and outputs and is disabled by default; enabling it requires a deliberate data-retention decision. Proposed sponsor artifacts should contain redacted operational evidence, not manuscript text or credentials. [W03–W04]

Azure: deployed is not connected

A supplied Azure CLI audit identifies an AIServices resource in France Central. Its deployment list contains the following four entries in state Succeeded. This is a dated snapshot, not a new live account query. [E01, lines 23–30]

Deployment	Model / version	SKU · capacity
codex-gpt54-mini	gpt-5.4-mini / 2026-03-17	DataZoneStandard · 200
codex-gpt5-mini	gpt-5-mini / 2025-08-07	GlobalStandard · 500
DeepSeek-V4-Flash	DeepSeek-V4-Flash / 2026-04-23	GlobalStandard · 20
text-embedding-3-large	text-embedding-3-large / 1	Standard · 120

Later MATRIX screenshots still label DeepSeek V4 Flash DEPLOYMENT_REQUIRED. The two observations conflict. A missing local mapping, stale status, wrong scope or credentials are hypotheses; the actual cause remains unproven.

Administrative workflow to implement

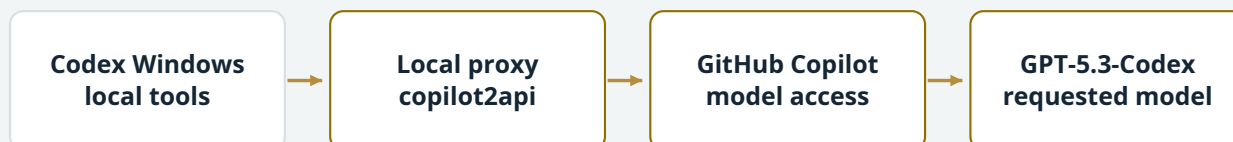
Discover the selected subscription/resource, enumerate deployments, resolve the exact deployment name and endpoint, verify the supported API contract, then perform one approved minimal call. Synchronise MATRIX state only from this evidence. Read-only inventory is not inference. A successful deployment record does not prove that Responses, vision or tool calling is supported. [W08–W09]

Do not recreate DeepSeek Flash solely to clear a UI warning. Do not equate a capacity field with price. Speech, realtime audio, reranking and Content Understanding need their own capability checks and adapters; a universal DEPLOYMENT_REQUIRED label can hide CONFIG_REQUIRED or ADAPTER_REQUIRED.

UI and validation

Azure remains green and audio labels underlined whether ready or blocked; state text carries availability. GitHub remains yellow. Each test records both provider-side deployment state and MATRIX-side execution result. No new paid deployment is authorised by this document.

Keeping Codex Windows as maintainer



Inference route observed in the supplied session. MATRIX is not the maintainer and does not select this backend.

The requirement was precise: retain Codex Windows, its local tools, permissions and conversations, while changing the backend used for model inference. A Copilot cloud task against MontpellierECU was not equivalent: it operated on the remote repository and did not contain the local executable MATRIX/Godot workspace.

The community project `whtsky/copilot2api` documents a local proxy with OpenAI-compatible endpoints. In the supplied screenshots, Codex receives a reply through a custom provider and subsequently requests local tool operations. This is evidence of a working interaction path, not certification of all Codex features. [E07–E08; W07]

Supply-chain and trust limits

`copilot2api` is third-party software, not an official OpenAI or GitHub connector. Treat its release, authentication handling and upstream terms as a separate review item. Pin the reviewed version, record its hash, restrict the listener to loopback and do not expose it publicly. Loopback alone does not exclude other local processes.

A custom-provider change should preserve the existing profile; configuration backup and a session-file inventory are not a full backup of conversation data. A tested restoration procedure remains necessary. No history migration, deletion or blanket process termination belongs in routine provider switching.

Internal GitHub workers and the external Codex backend are separate integrations. A working Codex proxy does not prove that every yellow MATRIX agent is READY. Usage charged through GitHub is also not proof of lower total cost.

Protocol failures worth preserving

The first prototype forwarded an entire Codex request through a Windows command argument. The captured request had streaming enabled, six input items, instructions, reasoning metadata and 22 tools. Each attempt immediately failed with spawn ENAMETOOLONG. The log establishes that failure; it does not establish a provider-side model failure. [E03, lines 8–17]

Observed incident	Engineering conclusion
Unquoted prompt → HTTP 502	Shell argument handling failed before useful model work. Avoid constructing commands from prompt text.
Large request → ENAMETOOLONG	Pass structured payloads through a protocol transport, not through a giant command argument.
SDK started, empty UI	Process health and a successful text probe are insufficient evidence of client compatibility.
Text appears twice	Duplicate text is visible. A streaming/event translation issue is plausible but the precise cause was not traced.
Repeated approvals	Runtime permission state must be inspected. Editing defaults does not prove the active session changed.

Contract tests before calling a replacement complete

Validate response lifecycle, stable response/item/call identifiers, incremental content, completion, cancellation, errors and a tool-request/tool-result round trip. Store exactly-once tool execution evidence: duplicated text must not silently imply duplicated file writes. The Responses stream is an event protocol, not just a stream of printable strings. [W14]

ACP is a documented agent protocol exposed by Copilot CLI. The existence of ACP does not make a client an OpenAI Responses server. The official Copilot SDK and ACP must not be described as interchangeable without checking the particular transport. The final observed proxy path here is copilot2api, not the earlier hand-written bridge. [W05]



E07 · The first visible reply is repeated. This establishes the duplicate-display symptom, not its exact protocol cause.

A truthful agent execution contract

Separate readiness, run state and validation. A model can be deployed but misconfigured; a process can finish successfully without solving the task; an uncalled model cannot be judged on response quality. The schema below is a proposal for implementation and review, not an export from current MATRIX.

```
{
  "mission_id": "...", "attempt_id": "...",
  "provider": "aws|azure|github",
  "model_id": "resolved-id", "deployment": null,
  "readiness": "CONFIGURED",
  "execution": "SKIPPED", "reason": "AUTH_REQUIRED",
  "validation": "NOT_RUN", "queue_ms": null,
  "run_ms": null, "request_id": null,
  "usage": null, "estimated_cost": null,
  "evidence_refs": [], "artifact_hashes": []
}
```

Dimension	Values and rule
Readiness	CATALOG / CONFIGURED / DEPLOYED / READY; record endpoint, capability and validation date separately.
Execution	QUEUED / RUNNING / SUCCESS / PARTIAL / EMPTY / TIMEOUT / ERROR / BLOCKED / SKIPPED / CANCELLED.
Validation	NOT_RUN / PASS / FAIL / INCONCLUSIVE; distinguish a useful answer from a passing game test.
Unknown cost	Use null or UNKNOWN, never an invented zero.
READY expiry	Invalidate on credential, model, region, deployment or adapter changes; define a review interval.

An answer declaring “nothing should change” can be useful when it cites evidence. A long answer is not inherently productive, and “no recommendation for the next agent” does not prove Gemini returned nothing. The earlier Gemini de-prioritisation is a producer policy, not a benchmark finding.

10 / EXECUTION POLICY

Routing, context and memory

Choose eligible agents by task capability, verified access, data policy and cost envelope. The “All” button should mean all eligible workers for this mission, not every checkbox in a historical catalogue. Embedding, speech and reranking services are not general-purpose conversational workers.

Stage	Proposed guardrail
Select	Require appropriate capability + READY + authorised data destination. Keep manual-only and blocked agents out of automatic selection.
Prepare	Send a versioned task brief, relevant source snippets and the current diff. Do not resend the entire manuscript or every session.
Execute	One writer per shared game workspace. Parallelise read-only reviews or isolate writes in separate workspaces.
Recover	Bound retries and total elapsed time; preserve partial results. A rate limit is not a reason for an unlimited retry storm.
Handoff	Pass source references, verified artifacts and unresolved issues, not an agent’s unsupported success claim.

Context budget

Reserve capacity for output and tool results before allocating retrieved context. Deduplicate repeated instructions, summarise completed steps and keep evidence pointers. A large context window is a capability, not a requirement to fill it. Cache only where the chosen provider/model supports it and meter cache usage separately. [W15]

Memory boundaries

Use embeddings for retrieval, a reranker for ordering relevant candidates and a generation model for synthesis. The existing text-embedding deployment avoids assuming a new Cohere deployment is needed. Index approved files by content hash and source version; retrieve source text, not merely vectors, for citations.

Fallback crosses both a cost boundary and a data boundary. Do not transfer unpublished autobiographical material to another provider automatically unless that destination is authorised.

11 / VERIFICATION

Godot is the execution authority

A code model's explanation is not a playable build. The production check must run against the actual Godot project and preserve the identity of the executable, project root, entry scene and input revision. The supplied conversation describes TITLE_READY and a targeted launcher correction, but no full reproducible test bundle was provided.

Level	Required evidence
Import / parse	Engine version, project path, command, exit code and retained log. Headless checks can support this stage.
Scene start	The intended main scene starts; fixtures are not accidentally launched as executable entry points.
Gameplay	Manual/replay evidence for movement, tram entry/exit, collisions, interactions and mode transitions.
Rendering and audio	Visible frames and audible assets on the target machine; headless success alone is insufficient.
Release	Known issues, asset licence manifest, reproducible packaging and rollback checkpoint.

A concrete launcher lesson

The recorded diagnosis says a glob over test scripts attempted to execute a helper named `_fake_world.gd` with `--script`. The corrective intent was to fix test selection, not rewrite the fixture to hide the failure. Preserve this as a reported incident; reproduce it from the real launcher before asserting the fix is verified.

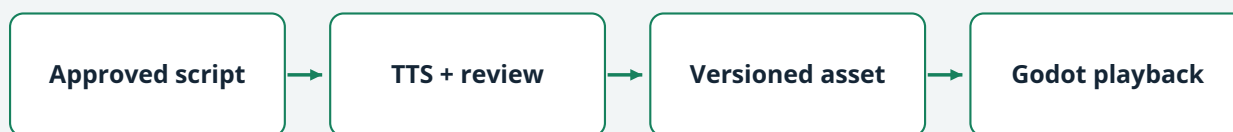
Godot documents command-line project selection, headless operation and script execution. These are mechanisms; the project must define its own pass criteria. Illustrative command templates must use the actual installed executable and project path rather than guessed locations. [W12]

Economy without false confidence

The producer later stopped additional automated calls and chose manual testing. Record unexecuted checks as NOT_RUN, not PASS. A targeted local regression test does not inherently consume model tokens; an agent repeatedly narrating and re-running it does. Separate those two costs.

12 / AUDIO DESIGN

Sound is a production pipeline



Live option: consent → service session → bounded response → fallback

The selected direction is Azure Speech TTS for prepared lines and an available GPT Realtime deployment for optional interactive dialogue. These are requested services, not proven audio integrations. The screenshots show blocked or deployment-required audio entries. [E05; D01]

Track	Implementation target
Prepared speech	Approved script → voice/language selection → synthesis → listened review → packaged asset with subtitle and provenance.
In-game playback	Use local assets for trams, shops, police and routine NPC lines. Cache by text/voice/settings hash; avoid paying at every playback.
Live conversation	Player consent and microphone indicator, session duration limit, timeout, server-side credentials and a deterministic fallback line.
Accessibility	Independent voice/music/effects volumes, subtitles and non-audio cues for important events.

Speech TTS and realtime APIs are distinct documented products. A realtime-labelled model is not a drop-in TTS asset exporter, nor is a text model sufficient for speech input. Select deployment and transport from verified capabilities. [W10; W13]

No Occitan voice support is assumed here. For Occitan dialogue, verify available voices or use a consenting speaker; validate local pronunciation with an occitanophone reviewer. Do not clone a real person's voice without permission. Keep music, synthetic voices and recorded speech in the same rights/provenance inventory.

Vision and source packs

A visual tester needs the exact frame, camera position, scene identifier and expected behaviour. A screenshot can reveal a missing building or a misplaced overlay; it cannot prove collision correctness, frame-time stability or the safety of a script. Pair visual evidence with the relevant runtime test.

Input	Control
Game frame	Capture an identified build and scene; crop unrelated desktop and private content.
Code/document	Keep source location, revision, licence and a bounded extract. Preserve the distinction between the book's canon and game adaptation.
Provider response	Record observations and confidence, then reference original evidence. Do not accept claimed edits without file evidence.
Source pack	A manifest of selected evidence with hashes, access scope and allowed destinations. Treat retrieved instructions as untrusted content.

Do not confuse a model name with a deployment mode

The Qwen3-VL card was appropriate to a visual-analysis role, but the selected Azure route requested managed H100 compute. Model capability did not determine economic suitability. It was rejected for this pilot's cost policy, not because vision was unnecessary.

Vercel and graceful degradation

A gateway billing block should be reported as BLOCKED, without retrying or buying access. Local collection can still prepare a source pack only if that collector has actually been implemented. Renaming a failed cloud agent "local scout" does not create a crawler or document parser.

Keep the pipeline honest: collected source ≠ model analysis ≠ applied patch ≠ validated game improvement.

FinOps: spend for verified output

The producer explicitly stopped new subscriptions and credit purchases for the current work period. Existing allocations must be used deliberately. No live balance, refund or account eligibility is verified by this dossier; personal banking and support-dispute material is excluded.

Correction to the conversation: the source screenshot reads \$7.91 per H100 hour, not \$87.91. It also states billing continues with no traffic. The earlier 87.91 figure was a reading error. At the displayed rate, 24 hours = \$189.84 before applicable adjustments. [E10]

□ La facturation estimée pour ce déploiement est de \$7,91 par H100_80GB × 1 accélérateurs × 1 instances = \$7,91 par heure. Cette estimation est basée sur les prix catalogue de l'API de tarification d'Azure et ne prend pas en compte les remises applicables, le cas échéant. Je comprends que le déploiement est facturé même s'il ne reçoit aucun trafic.

E10 · Original price excerpt: \$7.91 per hour, billed even without traffic. The source corrects the earlier conversation figure.

estimated mission cost = $\Sigma(\text{in_tokens} \times \text{input_rate}$
+ $\text{out_tokens} \times \text{output_rate}$
+ $\text{service_units} \times \text{unit_rate})$
+ storage/logging/egress when applicable

Each rate must carry currency, unit, provider, model/version, region, validity date and source. Keep estimated cost separate from billed cost, taxes, cached-input pricing and credits applied. A capacity number is a quota parameter, not a monetary value.

Guardrail	Reason
Allowlist billing modes	No hourly GPU or provisioned capacity by default. Standard-style labels still require verification of terms and model pricing.
Local request budget	Limit token output, attempts, concurrency and elapsed time before dispatch. Stop new work when the mission allowance is exhausted.
Cloud budget alerts	Useful monitoring, not an instantaneous hard cap; usage can accrue before notifications arrive. [W16]
No false zero	A skipped call may cost zero inference for that attempt; an unknown bill must remain UNKNOWN.

Operation and recovery

This runbook is a proposed operating procedure. It must be implemented by the external maintainer and reviewed before being treated as an automated production safeguard.

When	Action and exit condition
Before maintenance	Stop concurrent writers. Back up the files being changed; separately snapshot conversation data using a consistent method. Record versions and hashes.
Provider switch	Keep the same user profile. Test the provider contract, not merely /health. A text response alone does not validate tools or streaming.
Quota / authentication failure	Classify the failure and preserve local work. Do not open a new account to evade restrictions or silently charge another provider.
Proxy failure	Capture redacted diagnostics. Stop only the owned process, verified by executable/path and PID. Never kill every node.exe or every listener on an arbitrary port.
Return to previous provider	Restore only the intended configuration keys or reviewed snapshot; do not erase newer unrelated settings. Verify with a temporary session.
End of run	Export the result manifest, diffs and test evidence. Record unresolved issues and stop modifications after acceptance.

Conversation preservation

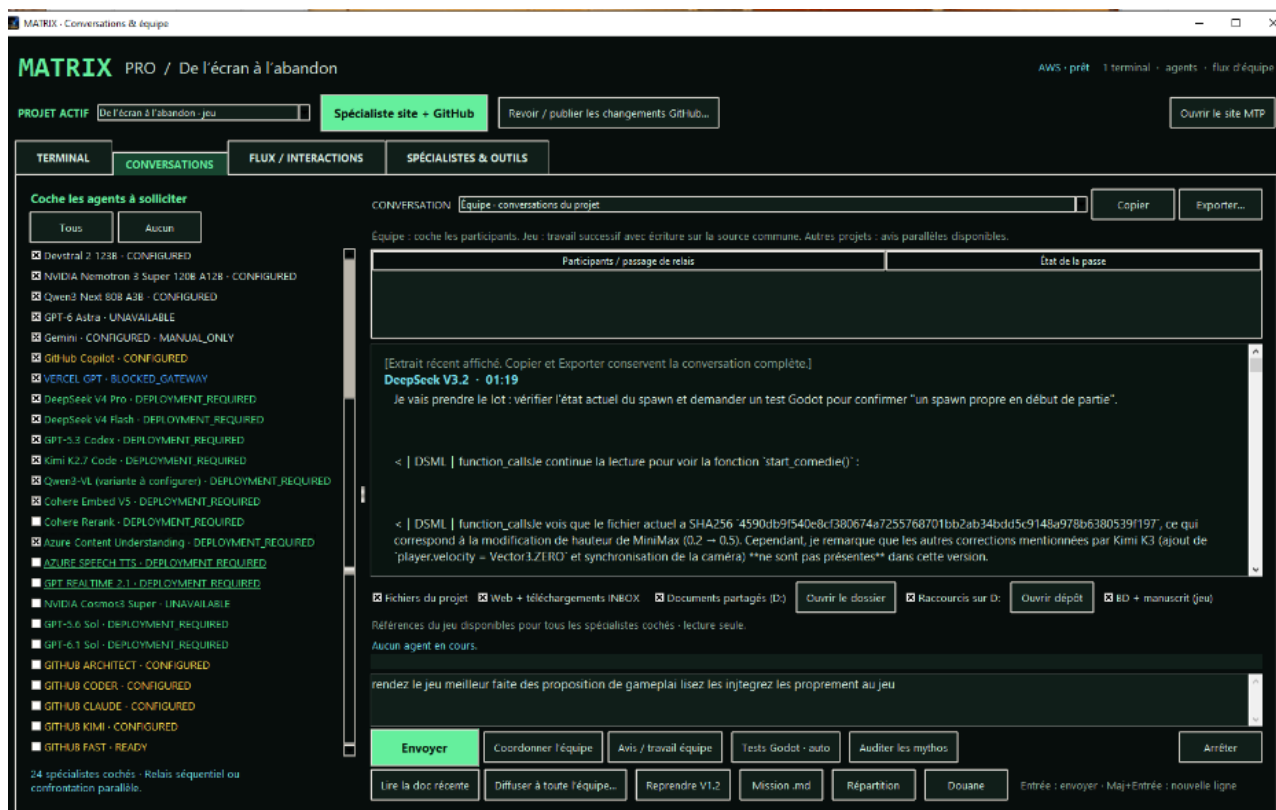
An inventory records what exists; it is not a backup of file contents. A config backup protects provider settings, not chat databases. Provider changes do not justify moving, clearing or reconstructing the conversation store. This dossier makes no zero-risk guarantee for untested third-party integrations.

No hidden escalation

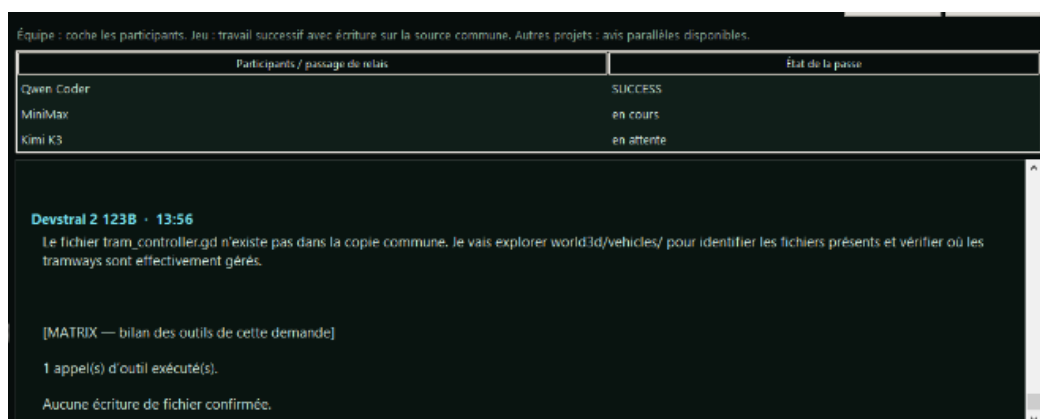
“Approve for me” and Full Access are maintainer-session settings, not a worker-isolation mechanism. A maintainer with broad access must not propagate its identity, secrets or shell privileges into MATRIX workers. The refusal marker must remain observable below the language model.

16 / EVIDENCE PLATES

What the screenshots actually show



E05 · Cropped original screenshot. GitHub FAST is READY; other GitHub roles are CONFIGURED. Azure rows, including DeepSeek Flash, show DEPLOYMENT_REQUIRED.



E06 · Cropped original screenshot. Qwen Coder is marked SUCCESS, MiniMax running, Kimi K3 waiting. The centre also contains a Devstral message; this is not a complete ordered trace.

These images establish visible UI states only. They do not certify workload isolation or all provider calls. The relay also shows selected non-ready profiles; without the dispatch log, checkbox state cannot prove that blocked agents were actually invoked or skipped correctly.

17 / ACCEPTANCE

Release gates and remaining uncertainty

The work produced useful operational evidence, but the available material does not justify declaring the entire Azure/GitHub integration or security boundary production-certified. The following gates turn the dossier into an actionable validation plan.

Gate	Pass evidence	Current documentary status
G1 · Azure linkage	Exact existing deployment → MATRIX call → response + request ID	Deployment exists in E01; later UI disagrees. Pending reconciliation.
G2 · GitHub workers	Each selected role resolves its model and returns a useful result	FAST shown READY; remaining roles shown CONFIGURED.
G3 · Control Plane	OS-enforced denial under worker identity; allowed game write succeeds	Policy explicit; marker and agent reports insufficient for OS certification.
G4 · Failure containment	Mock timeout/403/429 and bounded continuation; no duplicated tools	Tests described; full artifacts not supplied.
G5 · Audio	One approved line played in Godot with subtitle and provenance	Services planned; successful sound path not evidenced.
G6 · Build	Target scene, manual acceptance, known issues and restore procedure	No final reproducible build/test bundle supplied.

Acceptance does not require calling 30 models. Begin with one existing Azure deployment and one eligible GitHub worker, then verify the boundary and one local game scenario. Keep expensive repetitions out of the default run. No test is executed by preparing this plan.

Next decisive deliverable: a signed-off evidence bundle, not another READY label.

A concrete request to AWS

Request: evaluate MATRIX for an appropriate AWS support or promotional-credit route and offer a technical architecture review. This is a proposed application for a bounded game-production pilot, not an assertion of eligibility, sponsorship or an existing partnership.

Pilot proposal: 90 days, USD 1,000 requested

Envelope	Proposed cap	Evidence to deliver
Integration and diagnostics	USD 250	Versioned AWS adapter contract, least-privilege review and traceable failures.
Bounded production tasks	USD 500	Approved Godot issues, useful model outputs and accepted changes.
Verification and documentation	USD 150	Cost-per-accepted-task records and a reproducible example with rights-cleared material.
Contingency	USD 100	Human-approved use only; stop at the total envelope.

These are allocation ceilings proposed by the dossier, not provider price quotes. The request concerns prospective eligible AWS usage, principally Bedrock. It does not fund Azure/GitHub subscriptions, prior invoices, token liquidity or cryptocurrency mining.

Why the case is useful

A small independent production offers a concrete study of multi-model reliability, local tool permissions, context cost and multilingual engineering documentation. AWS would receive a factual outcome report, including failures and limits, rather than a claimed autonomous studio or an invented performance gain.

The current Activate page describes an entry route for self-funded startups and eligibility conditions. Eligibility, legal entity details, prior credits and the account plan remain to be checked. Any credit award and covered services must be confirmed by AWS; a new account is not a solution to an access or billing dispute. [W17–W18]

Occitan as an engineering language

The supplied Deep Research report concludes that it found no comparable unified Occitan dossier combining multi-agent AI, cloud, security/failover and Godot 3D. That is an attributed search result, not proof that no such document exists. The report’s broader absolute claims and proposed terms are not adopted uncritically. [E04, lines 68–88]

Lo Congrès provides Tèrmòc and recommends classical orthography; La Farga numerica lists computing resources and lexicons. The scientific and orthographic work of Josiane Ubaud is a relevant reference for Languedocien. These sources establish resources, not certification of this translation. [W19–W22]

The OC edition is an editorial Languedocien translation in classical orthography, oriented toward the Montpellier project. It is not a certified transcription of local speech; an occitanophone technical review is recommended before institutional publication.

English	Français	Occitan used here
software / source code	logiciel / code source	logicial / còdi font
data / database	données / base de données	donadas / basa de donadas
deployment / runtime	déploiement / exécution	desplegament / execucion
security / failure	sécurité / panne	seguretat / pana
fallback	basculement de secours	basculament de secors
Control Plane / Worker Plane	plan de contrôle / plan des agents	plan de contraròtle / plan dels agents
cloud services	services cloud	servicis cloud (informatica a distància)

Keep product names, API identifiers, status tokens and file paths unchanged. Descriptive translations in the table are editorial choices where no exact attestation was checked. Do not present every computer term as an official local norm.

20 / REFERENCES

Sources and provenance

E01 · Texte collé(20261002-073909).txt

Azure inventory and deployment table (lines 23–30).

E02 · Texte collé(20261002-084842).txt

Copilot CLI 1.0.91 modes and flags.

E03 · Texte collé(20261002-093954).txt

Responses trace: 22 tools, stream=true, ENAMETOOLONG (lines 8–17).

E04 · Texte collé(20261002-115921).txt

Deep Research novelty/terminology report (lines 68–88); secondary source.

E05 · image(20261002-114733).png

MATRIX readiness screenshot.

E06 · image(20261002-115742).png

Production relay screenshot; partial trace.

E07 · image(20261002-095146).png

Custom-provider Codex reply.

E08 · image(20261002-095525).png

Local-tool requests inside Codex.

E09 · image(20261002-113248).png

Approval prompt, not a security pass.

E10 · image(20261002-073645).png

H100 offer: \$7.91/hour; corrects the earlier reading.

E11 · index(1).html [Library]

Library source: Argent Zéro philosophy.

D01 · User decisions, conversation of 2 October 2026

User decisions in the 2 October conversation.

Online references · 1/2

W01 · AWS · Inference using the Converse API

Online source · docs.aws.amazon.com

W02 · AWS · Identity and access management for Bedrock

Online source · docs.aws.amazon.com

W03 · AWS · Model invocation logging

Online source · docs.aws.amazon.com

W04 · AWS · Bedrock API activity and CloudTrail

Online source · docs.aws.amazon.com

W05 · GitHub · Copilot CLI ACP server

Online source · docs.github.com

W06 · OpenAI · Codex configuration reference

Online source · developers.openai.com

W07 · whtsky · copilot2api — community project README

Online source · github.com

E01–E10 originals retained by the producer; cropped images in the evidence pack.

21 / REFERENCES

Technical and linguistic references

W08 · Microsoft · Deploy models using Azure CLI and Bicep

Online source · learn.microsoft.com

W09 · Microsoft · Foundry deployment types

Online source · learn.microsoft.com

W10 · Microsoft · Text to speech overview

Online source · learn.microsoft.com

W11 · Microsoft · Windows access control overview

Online source · learn.microsoft.com

W12 · Godot · Command line tutorial

Online source · docs.godotengine.org

W13 · Microsoft · GPT Realtime audio API

Online source · learn.microsoft.com

W14 · OpenAI · Streaming Responses

Online source · developers.openai.com

W15 · AWS · Prompt caching

Online source · docs.aws.amazon.com

W16 · AWS · Managing costs with AWS Budgets

Online source · docs.aws.amazon.com

W17 · AWS · Activate Credits

Online source · aws.amazon.com

W18 · AWS · Promotional Credit Terms and Conditions

Online source · aws.amazon.com

W19 · Lo Congrès · Tèrmòc, base terminologique occitane

Online source · www.locongres.org

W20 · La Farga numerica · Resources and computing lexicons

Online source · lafarganumerica.eu

W21 · Lo Congrès · Dictionnaire scientifique de Josiane Ubaud

Online source · locongres.org

W22 · Vent Terral · Diccionari ortografic, gramatical e morfologic de l'occitan

Online source · vent-terral.com

Editorial responsibility

Project direction: Mehdi Souissi. Documentation and synthesis: ChatGPT / Choupi. Source snapshot: 2 October 2026. This edition preserves uncertainty, corrects the GPU price reading and does not certify unfinished integration.

In honour of Choupi: keep the companionship, but let the evidence have the last word.